

The role of venture capital in the creation of public companies

Evidence from the going-public process*

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We examine an exhaustive set of initial public offerings (IPOs) by venture-capital-backed companies between 1978 and 1987. We find that venture capitalists specialize their investments in firms to provide intensive monitoring services. Consistent with their monitoring role, the venture capitalists take concentrated equity positions, maintain their investment beyond the IPO, and serve on the boards of their portfolio firms. The quality of their monitoring services appears to be recognized by capital markets through lower underpricing for IPOs with better monitors.

1. Introduction

Venture capitalists usually take concentrated equity positions in the companies they fund and exercise significant influence on management. Their active involvement in their portfolio companies suggests similarities to the

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large stockholders and 'active investors' that have attracted attention in the corporate control literature [see, for instance, Holderness and Sheehan (1985, 1988), Mikkelsen and Ruback (1985), and Shleifer and Vishny (1986)]. These active investors have the potential to play important roles in monitoring and reorganizations [see Jensen (1989)]. Although there are clear indications that ownership concentration and firm behavior are related, Jensen and Warner (1988) conclude that the evidence on the role of large blockholders, including their monitoring function, is mixed.

Venture capitalists are similar to leveraged-buyout (LBO) specialists, who appear to contribute to substantial improvements in performance in corporations that go private [see Kaplan (1989), Muscarella and Vetsuypens (1990), and Smith (1990)]. Like LBO specialists, venture-capital providers are active in financing and monitoring their portfolio companies. LBO specialists and venture capitalists invest in different kinds of firms, however: LBO promoters usually invest in mature companies with predictable cash flows, whereas venture capitalists focus on young and high-risk entrepreneurial ventures. The venture-capital market therefore provides a useful alternative setting in which to explore the role of active investors with concentrated ownership claims.

In this paper, we use data on initial public offerings (IPOs) to explore the nature of venture-capital investment. Venture capitalists invest in privately held companies that are not visible to the public. At the time of an IPO, however, the offering prospectus identifies venture-capital investors, explains their investment position, and gives some details of their activity within the IPO firm. IPO data, therefore, provide a unique opportunity to examine venture capitalists' activities.

We study an exhaustive set of 433 IPOs by venture-capital-backed companies over the period from 1978 through 1987. The data suggest that venture capitalists take economically significant equity positions in their portfolio firms. The lead venture capitalist averages a 19% ownership stake at the time of the IPO, and the aggregate holdings by all venture capitalists represent, on average, 34% of the outstanding shares. Venture capitalists participate directly in the governance of their portfolio companies. They hold about one-third of the board seats and have assisted these companies for about half of their corporate lives before the IPO.

Many venture capitalists are frequent participants in the capital markets. Our findings support the notion that a venture capitalist's expertise and experience in monitoring investments can send important signals to investors at the time of an IPO. Consistent with Sahlman's (1990) view of the nature of venture capitalists' contracts with their investors and with entrepreneurs, we find that venture capitalists are willing to bind themselves to the value of a new issue by maintaining their equity positions beyond the IPO. Furthermore, the capital markets appear to recognize the quality of venture-capital

monitoring services by requiring less underpricing for issues with higher-quality monitors.

This paper is the first to examine in detail the empirical regularities in venture-capital investment across a large sample of observations.¹ It complements the work by Sahlman (1990), who describes how venture capitalists structure contracts to minimize agency conflicts with entrepreneurs and with investors in venture-capital funds. Because little is known about venture capital, this paper is as much an exploratory analysis as a test of specific hypotheses on the role of venture capitalists.

The plan of the paper is as follows. Section 2 gives an overview of the venture-capital industry. Our data set of venture-capital-backed companies going public is presented in section 3, where we outline our data collection procedure and describe key attributes of the sample. In section 4, we present empirical evidence of venture capitalists' active participation in the sample firms and analyze the effects of venture-capital quality on the IPO. Conclusions are presented in section 5.

2. Venture capital

The venture-capital industry provides risk capital to companies that offer high potential returns. The industry grew rapidly in the 1980s: the *Venture Capital Journal* reports that new capital commitments by independent venture capitalists (as opposed to the venture-capital subsidiaries of commercial banks, corporations, and investment banks) totalled \$4.18 billion in 1987, compared with \$218 million in 1978. At year-end 1988, 658 independent venture-capital providers managed an aggregate of \$31 billion, compared with 211 venture capitalists managing \$2.5 billion in 1977 (*Venture Capital Journal*, October 1982 and March 1989). Most venture-capital portfolios are relatively small. The *Venture Capital Journal* data show that in 1988, 359 venture capitalists (54.6% of the total) managed in aggregate only \$3 billion (9.7% of the capital under management), while 32 venture capitalists (4.9% of the total) managed a combined \$10.6 billion (34.1% of the capital under management).

Venture capitalists are typically active investors who try to 'add value through ongoing longer-term involvement with continuing business development' (*Venture Capital Journal*, March 1987). They often invest through syndicates in which one or more venture capitalists take the lead role in working with the portfolio company. Venture capitalists often participate directly in management; typically one or more serve on the company's board of directors. A venture-capital provider often specializes in a particular

¹To our knowledge, the only other large-sample study of venture capital in new issues is Megginson and Weiss (1990).

industry [see Warne (1988)] and uses its industry contacts to help the company recruit key employees, to assist in production, to line up suppliers, and to develop customer relations. Gladstone (1989) reports that some venture capitalists maintain consulting staffs that participate actively in the management of portfolio companies.

Venture capitalists usually provide financing in stages corresponding to significant developments in the life of the venture, for instance, the formulation of a business plan, first production, and so on [see Sahlman (1988) and Kane (1989)]. Staged financing allows the venture capitalists periodically to revalue an investment and abandon it if the expected net present value turns negative. Sahlman (1988) shows that staged financing solves agency conflicts between the venture capitalist and the entrepreneur because the entrepreneur has a stronger incentive to create value than if all needed capital were provided at once. The company becomes progressively less risky as it develops, and the venture capitalist will accept a proportionately smaller incremental equity position for a given amount of investment at each successive stage [see also Chan, Siegel, and Thakor (1987)]. In bonding the entrepreneur to create value, staged financing is analogous to the heavy interest burden created by leveraged buyouts, which can generate similar incentives to maximize value [see Jensen (1986)].

Like LBO specialists, venture capitalists eventually want to liquidate their ownership position. As DeAngelo and DeAngelo (1987) point out, the comparative advantage of the LBO specialist is its ability to oversee the efficient restructuring of operations. Once this task has been completed, the specialist's marginal productivity will be higher if it redeploys its resources elsewhere. Similarly, because venture capitalists' unique skill is the guidance of entrepreneurial ventures in early stages of growth, they will rationally leave their portfolio companies when those companies become mature.

Gladstone (1989) describes six ways of concluding a venture capital investment: (1) sale of the company's shares in a public offering, (2) sale of shares to another company, (3) repurchase of the shares by the company, (4) sale of shares to another investor, (5) reorganization of the company, and (6) liquidation of the company. We explore the first, which appears to be the alternative most frequently chosen. Among 544 venture-capital investments made between 1970 and 1982, 35% used an IPO, whereas the next most common outcome, acquisition by another company, accounted for only 22% of the investments, according to the *Venture Capital Journal*.² Almost all (96%) of the IPOs, but only 59% of the acquisitions, provided positive returns for the venture capitalists. Hence, there may be systematic differ-

²See 'Exiting: New Patterns in the 1980s', *Venture Capital Journal*, August 1988, pp. 12-16.

ences in the characteristics of companies and venture capitalists that are able to choose the IPO alternative.

3. Sample-selection procedure

We obtained two samples of initial public offerings: 433 IPOs with venture-capital backing and 1,123 IPOs without such backing. The venture-backed sample includes detailed data from the offering prospectus for each IPO on ownership, board membership, and identity of investors. This data set permits us to examine the roles played by the venture capitalists in their portfolio firms. The second sample, without venture-capital backing, was obtained to compare the characteristics of offerings with and without venture capital. This comparison focuses on potential differences in offering characteristics (such as initial returns or the quality of the underwriter) that might be associated with venture capitalists' role as monitors and active investors.

Venture-backed offerings are initially identified using the lists of new issues with venture-capital participation that appear annually in the *Venture Capital Journal*, published by Venture Economics, Inc. Venture Economics maintains a database of venture-capital providers and of the companies in their investment portfolios. Over the period 1978–1987, Venture Economics reported on 539 IPOs made by companies that had received venture-capital investment.³

To assess the participation and ownership of venture capitalists, we examined the 'Management' and 'Principal and Selling Shareholders' sections in each IPO prospectus. The 'Management' section identifies the offering firm's top executives and directors. Usually, board members who represent venture capitalists are designated as such. The 'Principal and Selling Shareholders' section lists equity ownership before and after the IPO for all officers, directors, and 5% shareholders, and frequently the holdings of investors with smaller fractional ownership.

We cross-referenced the stockholders listed in the prospectus against *Pratt's Guide to Venture Capital Sources* (hereinafter *Pratt's Guide*), an annual directory of venture-capital practitioners published by Venture Economics. For each shareholder listed in *Pratt's Guide* who held at least 1% ownership in the issuing firm before the IPO, we recorded the percentage equity ownership and noted any board membership. To avoid a survivorship bias, we used both current and older issues of *Pratt's Guide*. For most IPOs, this initial search identified at least one stockholder listed as a venture-capital provider in *Pratt's Guide*. Some stockholders described in the prospectus as

³Venture Economics excludes IPOs with offer prices below \$5.00 or offer sizes below \$3 million.

venture-capital providers were not listed in *Pratt's Guide*, however (possibly because of parent/subsidiary links among venture-capital affiliates that we could not recognize). We also examined the 'Recent IPOs' and 'IPOs in Registration' sections of the *Venture Capital Journal*, which list investors that have provided venture capital to issuing companies. Any additional venture-capital providers discovered in these lists were added to our sample.⁴ From the original 539 venture-backed IPOs listed in the *Venture Capital Journal*, 106 were eliminated for the following reasons: (1) we were unable to obtain IPO prospectuses for some of the companies; (2) we eliminated offerings of limited partnership units and offerings of reverse LBOs (i.e., IPOs of companies that were acquired by management and the venture capitalist in a leveraged buyout of the predecessor public company or in a division buyout); and (3) we excluded companies for which no venture capitalists were identified as shareholders in the IPO prospectus (even though the *Venture Capital Journal* listed the company as being venture-capital-backed).

We also compiled a control sample of IPOs without venture-capital backing over the same period. For the years 1983 through 1987, this control sample consists of all non-venture-capital-backed IPOs that were announced on the *Dow Jones News Wire* or in the *Wall Street Journal*. Small offerings made by little-known companies will be underrepresented in this sample, since editors are less likely to devote *Journal* or *Wire* space to such issues. For the period 1978–1982, we used a subset of the Chalk and Peavy (1990) IPO data set compiled from *Going Public: The IPO Reporter*.

As table 1 shows, the annual number of venture-capital-backed IPOs and the amount of capital the IPOs raised (in nominal dollars) have increased over time. The second half of the sample period (1983–1987) accounts for three-quarters of the venture-capital-backed IPOs. The amounts of capital raised exceeded \$1 billion in both 1986 and 1987, up from only \$68 million in 1978. In total, the sample venture-capital-backed IPOs raised \$8.2 billion from 1978 through 1987. The total value of all the shares outstanding immediately after the IPO for these companies (not reported in table 1) exceeds \$36 billion.

⁴We assume the ownership information in the prospectus is accurate and ignore venture-capital providers listed in the *Venture Capital Journal* but not identified in the offering prospectus. The discrepancy between the *Venture Capital Journal* and the prospectus may be caused by a variety of factors: investors identified by the *Venture Capital Journal* (a) may have liquidated their holdings before the IPO, (b) may own too small a percentage to be listed separately in the prospectus, (c) may have invested in the issuing firm indirectly through a venture-capital firm that is itself listed in the prospectus, (d) may have provided nonequity capital that is not explicitly described in the 'Selling Shareholders' section of the prospectus. Another possibility is that the *Venture Capital Journal* list is simply incorrect. This last explanation appears to be the most likely, for we also found venture-capital investors listed in the IPO prospectus but not so identified in the *Venture Capital Journal*.

Table 1
Frequency distribution of venture-capital-backed and non-venture-capital-backed initial public offerings over the period 1978–1987.^a

Year	Number of IPOs		Venture-capital-backed IPOs as percentage of Total	Total capital raised in venture-capital-backed IPOs (in millions of \$)	Number of IPOs in Ibbotson, Sindelar, and Ritter (1988)	Venture-capital-backed IPOs as percentage of IPOs in Ibbotson, Sindelar, and Ritter (1988)
	Venture-capital-backed	Non-venture-capital-backed				
1978	6	3	9	\$ 68	42	14%
1979	4	2	6	34	103	4
1980	24	21	45	404	259	9
1981	50	61	111	517	438	11
1982	21	43	64	518	198	11
1983	101	284	385	2,525	848	12
1984	44	151	195	555	516	9
1985	35	125	160	643	507	7
1986	79	252	331	1,614	953	8
1987	69	181	250	1,331	630	11
1978–1987	433	1,123	1,556	\$8,209	4,494	10%

^aThe venture-capital-backed IPOs are all new issues with venture-capital participation as reported in the *Venture Capital Journal* for which we could verify the identity of venture capitalists against *Pratt's Guide to Venture Capital Sources*. Venture-backed offerings of limited partnership units and reverse LBOs are excluded. The non-venture-capital-backed issues are compiled from *Going Public: The IPO Reporter* for 1978–1982, and from *Wall Street Journal* and *Dow Jones News Wire* announcements for 1983–1987.

Over the period 1978–1987, venture-capital-backed IPOs represent about 28% of all IPOs in this study. Our sample of non-venture-capital-backed IPOs is not exhaustive, however. Accordingly, we also report in table 1 a frequency distribution of 4,494 IPOs tabulated in Ibbotson, Sindelar, and Ritter (1988). Their data are from Ritter (1984) for 1978–1982, *Going Public: The IPO Reporter* for 1983–1985, and *Venture* magazine for 1986–1987. Our 433 venture-capital-backed IPOs represent about 10% of this apparently comprehensive listing of IPOs. This proportion does not vary much over the ten-year period under study.

Table 2 shows that a total of 424 venture capitalists provided financing to the 433 venture-capital-backed IPO companies in our sample (panel A). Independent venture capitalists account for 349 (82%) of these investors, and venture-capital subsidiaries of underwriters, banks, industrial firms, and insurance companies constitute the remainder. Many IPOs were backed by more than one venture capitalist; the 433 issuers received 1,264 separate venture-capital investments, or roughly three per IPO firm on average. The participation by multiple venture capitalists contrasts with the case of LBO specialists, who rarely share equity or board membership in their portfolio companies with other buyout firms. The relatively small size of most venture capitalists' portfolios may explain this difference. Multiple venture-capital participation in a company may enable the company to achieve an efficient scale of operations while at the same time improving the efficiency of risk sharing. Another explanation for the participation of multiple venture capitalists is the benefit of additional monitoring. The presence of multiple venture capitalists allows the originating venture capitalist to obtain independent assessments about the likely success of the venture.

Venture capitalists tend to be young companies (table 2, panel B). Of the 210 venture-capital providers in our sample for which *Pratt's Guide* reported the year of founding, nearly half (46%) were established during the 1980s.

A small number of well-known venture-capital investors appear frequently in our sample. The five most active are: Hambrecht and Quist (40 investments), Citicorp Venture Capital (32), Kleiner, Perkins, Caufield, and Byers (30), Greylock (22), and T.A. Associates (22). Such venture capitalists become well-known among issuers, underwriters, and investors in the IPO market, and their success in bringing companies public becomes a selling point as they negotiate venture-capital contracts with entrepreneurs and solicit funds from prospective investors.

In their study of majority stockholders, Holderness and Sheehan (1988) point out that the organizational role of large-block shareholders could in principle be either value-increasing or value-decreasing. In contrast to majority blockholders in a single firm, the frequency of venture capitalists' investments suggests that their involvement in their portfolio firms is geared toward

Table 2
Descriptive statistics on venture-capital investors participating in 433 initial public offerings, 1978–1987.

Panel A: Frequency distribution of participation by 424 venture capital investors											
	Total	Independent venture capitalists	Venture- capital subsidiary of underwriter	Venture- capital subsidiary of bank	Venture- capital subsidiary of industrial company	Venture- capital subsidiary of insurance company					
Number of venture capitalists	424	349	16	24	26	9					
Percent of total	100.0%	82.3%	3.8%	5.7%	6.1%	2.1%					
Number of investments	1,264	937	131	108	57	31					
Percent of total	100.0%	74.1%	10.4%	8.5%	4.5%	2.5%					
Panel B: Frequency distribution of founding year of 210 venture capital investors											
	Before 1960	1960-69	1970-79	1980	1981	1982	1983	1984	1985	1986	1987
Number of venture capitalists	11	44	63	14	20	22	20	10	2	2	2
Percent of all venture capitalists reporting founding year	5.2%	21.0%	30.0%	6.7%	9.5%	10.5%	9.5%	4.8%	1.0%	1.0%	1.0%

creating value. Failure to do so would damage their ability to arrange subsequent deals.

4. Empirical results

In this section, we present evidence on the active participation by venture-capital investors in the sample firms. First, we contrast venture-backed IPOs with IPOs that have no venture backing. We then examine the role that venture-capital providers play as large shareholders and monitors in the IPO firms. We conclude by demonstrating that proxies for venture capitalists' monitoring skills are associated with reduced underpricing at the time of the IPO.

4.1. *IPOs with and without venture capital*

Table 3 shows that the venture capitalists in our sample tend to focus their investments in a subset of industries. Computer equipment, electrical and electronic components, instrumentation, and business services (which in-

Table 3
Frequency distribution of initial public offerings by two-digit SIC code, for venture-capital-backed and non-venture-capital-backed issues, 1978–1987.^a

SIC code	Industry description	Percentage of all venture-capital-backed IPOs	Percentage of all non-venture-capital-backed IPOs
28	Chemicals and allied products	4.7%	3.8%
35	Industrial and commercial machinery and computer equipment	18.9	5.2
36	Electronic and other electrical equipment and components, except computer equipment	15.5	7.0
38	Measuring, analyzing and controlling instruments, photographic, medical and optical goods; watches and clocks	7.6	5.1
50	Wholesale trade-durable goods	1.0	3.9
58	Eating and drinking places	1.8	3.7
59	Miscellaneous retail	0.2	3.0
67	Holding and other investment offices	1.6	5.0
73	Business services	21.8	13.3
80	Health services	5.2	3.5
	All other	21.7	46.5
		100.0%	100.0%

^aSIC codes listed are all those that account for at least 3% of either the venture-capital-backed or the non-venture-capital-backed IPOs.

cludes computer software) account for 63.8% of the venture-capital-backed IPOs, but only 30.6% of other IPOs. In *Pratt's Guide*, many venture-capital listings explicitly indicate formal industry preferences such as computers, electronics, and communications. Similarly, venture capitalists often list industries they 'will not consider'. Thus, venture capitalists tend to develop expertise in a relatively narrow set of industries. Their specialized industry

Table 4

Descriptive statistics on offer size, underwriter rank, aftermarket standard deviation, and initial return for venture-capital-backed and non-venture-capital-backed initial public offerings, 1978–1987.

Variable	Mean	Median	First quartile	Third quartile	Sample size
<i>Panel A: Non-venture-capital-backed IPOs</i>					
Offer size ^a	\$26.3MM	\$10.0MM	\$5.0MM	\$21.3MM	1,120
Underwriter rank ^b	6.19	7.00	5.00	8.00	926
Aftermarket std. dev. ^c	2.90%	2.53%	1.78%	3.64%	1,119
Initial return ^d	7.47%	1.29%	0.00%	9.59%	1,120
<i>Panel B: Venture-capital-backed IPOs</i>					
Offer size ^a	\$19.1MM ^e	\$13.5MM	\$8.0MM	\$22.5MM	429
Underwriter rank ^b	6.94 ^e	7.5	5.5	8.0	409
Aftermarket std. dev. ^c	3.19% ^e	2.91%	2.04%	3.87%	430
Initial return ^d	8.43%	2.78%	0.00%	11.69%	430
<i>Panel C: Non-venture-capital-backed IPOs publicly announced in 1983–1987</i>					
Offer size ^a	\$28.3MM	\$10.4MM	\$5.0MM	\$22.5MM	991
Underwriter rank ^b	6.38	7.0	5.0	8.0	797
Aftermarket std. dev. ^c	2.86%	2.50%	1.71%	3.55%	990
Initial return ^d	7.71%	1.47%	0.00%	9.21%	991
<i>Panel D: Venture-capital-backed IPOs publicly announced in 1983–1987</i>					
Offer size ^a	\$22.9MM	\$16.8MM	\$10.8MM	\$26.4MM	219
Underwriter rank ^b	7.19 ^f	7.5	6.0	8.0	211
Aftermarket std. dev. ^c	2.85%	2.66%	1.90%	3.61%	220
Initial return ^d	6.93%	2.08%	0.00%	10.00%	220

^aDefined as offer price times shares sold to the public, excluding overallotments.

^bDefined as the Carter and Manaster (1990) ordinal ranking of underwriters, where '9' indicates highly prestigious and '0' represents the lowest level of prestige.

^cComputed using aftermarket returns from trading day 2 through trading day 20. Aftermarket prices are taken from Standard & Poor's *Daily Stock Price Record*. Closing transaction prices are used whenever they are available; otherwise, bid prices are used.

^dDefined as percentage price change from offer price to closing price at the end of the first trading day.

^eSignificantly different at the 0.01 level from the corresponding number in panel A, using a *t*-test for differences in means.

^fSignificantly different at the 0.01 level from the corresponding number in panel C, using a *t*-test for differences in means.

knowledge, combined with their privileged position as corporate insiders, facilitates their monitoring role.

Offering characteristics of IPOs with and without venture-capital backing are compared in table 4, panels A and B. Venture-capital-backed IPOs have a larger median offering size than the non-venture-backed IPOs (\$13.5 million versus \$10.0 million), although the presence of several very large non-venture-capital-backed IPOs leads to a smaller mean size for venture-capital-backed offers (\$19.1 million versus \$26.2 million). Using the Carter and Manaster (1990) underwriter ranks, we find that the venture-capital-backed IPOs are sold by more prestigious underwriters.⁵ The aftermarket volatility is significantly greater for venture-capital-backed IPOs.⁶

The average initial-day return of venture-capital-backed issues (8.43%) is not significantly different from that of the IPOs without venture backing (7.47%). This finding does not support Baron's (1982) theory of the issuance process, in which underpricing is seen as a consequence of information asymmetries between issuer and underwriter. In tests not reported here, we examined two subsets of IPOs in which (1) a venture-capital investor was an affiliate of an investment banking firm (87 offers) or (2) a venture-capital provider was an affiliate of an investment banker that was itself the lead underwriter of the issue (19 IPOs). In neither case did we find evidence of reduced underpricing. These results are inconsistent with Baron (1982), but they confirm the findings of Muscarella and Vetsuypens (1989).

These comparisons should be interpreted with care, because our sample of non-venture-capital-backed IPOs was largely selected from *Wall Street Journal* announcements, and hence is likely to be biased toward larger, less underpriced offerings made through higher-tier underwriters. To provide a comparison free from selection bias, we examine in panels C and D IPOs (with and without venture-capital backing) that occurred after 1982 and were publicly announced in the *Wall Street Journal*. The only statistically significant difference revealed by this comparison is that venture-capital issues are brought to market by more prestigious underwriters (as was true for the full sample). The presence of venture-capital investors is not associated with significant differences in underpricing, offering size, or aftermarket volatility.

Venture capitalists sometimes claim that they help firms go public at higher price/earnings (P/E) ratios. Since the P/E ratio is discontinuous at zero earnings and since the use of earnings yield is commonplace in the literature [see, for instance, Basu (1983) or Reinganum (1981)], we looked at

⁵Because the Carter–Manaster underwriter-prestige measures are ordinal, the use of a t -test of differences in average underwriter ranks is only suggestive. A more appropriate chi-square test of association between underwriter-prestige classes ('high' versus 'low') and the presence or absence of venture capital, however, yielded qualitatively similar findings.

⁶Aftermarket volatility is defined as the standard deviation of daily returns computed over 19 days, starting one day after the offering.

Table 5

Comparison of pre-IPO earnings and earnings yields for venture-capital-backed versus non-venture-capital-backed IPOs in 1978–1987, by industry.^a

Sample ^b	Sample size	Average pre-IPO earnings yield			Chi-square test of association between sign of earnings in last fiscal year before IPO and presence of venture backing
		Venture-capital-backed IPOs	Non-venture-capital-backed IPOs	<i>t</i> -statistic for difference in means	
All IPOs	1,170	–1.54%	2.45%	4.31 ^c	49.2 ^c
SIC code 28	43	–1.13%	5.21%	1.49	4.7 ^d
SIC code 35	118	–3.53%	1.12%	1.49	0.7
SIC code 36	106	2.20%	2.75%	0.45	0.7
SIC code 38	65	–0.11%	2.67%	1.81 ^c	1.2
SIC code 50	34	4.93%	3.41%	–0.99	0.4
SIC code 58	32	0.75%	–2.78%	–0.68	0.5
SIC code 59	23	1.09%	2.89%	0.39	0.2
SIC code 67	27	–1.62%	–5.29%	–0.32	2.1
SIC code 73	117	–0.78%	0.99%	1.32	7.0 ^c
SIC code 80	44	–0.89%	0.40%	0.93	4.2 ^d

^aEarnings yield is defined as earnings per share in the last fiscal year before the IPO divided by offering price.

^bSee description of SIC codes in table 3.

^cSignificant at 0.01 level.

^dSignificant at 0.05 level.

^eSignificant at 0.10 level.

the firms' earnings yield (most recent year's earnings per share divided by offering price) rather than at the *P/E* ratio. Consistent with venture capitalists' claims, we find that the IPOs they back have a lower average earnings yield than do non-venture-backed issues (table 5, top row). Venture-backed IPOs as a group are also significantly more likely than non-venture-backed issues to have negative earnings in the fiscal year before the offering, according to a nonparametric chi-square test.

These results for the full sample seem to suggest that venture capitalists are able to bring companies public earlier than would otherwise be possible. On closer inspection, however, the differences seem to be related to the industry focus of venture capital rather than the presence of venture capital per se. When the data are analyzed within two-digit standard industrial classification (SIC) codes, earnings–yield differences between IPOs with and without venture capital largely vanish. Venture-backed issues show a significantly lower mean earnings yield only in SIC code 38 (instrumentation). The chi-square test of association between the sign of pre-IPO earnings and the

presence of venture backing is significant only for SIC codes 28 (chemicals), 73 (business services), and 80 (health services). We therefore conclude that venture capitalists specialize in industries in which it is relatively common for companies to go public with negative earnings, whether venture capital is involved or not.

In sum, the comparison of IPOs with and without venture-capital backing reveals two important differences. First, the issues with venture-capital backing tend to be specialized in a relatively narrow set of industries. This specialization is consistent with venture capitalists' claims that they invest in industries in which their expertise and their ability to monitor and guide are greatest. Second, issues with venture backing are offered through higher-quality underwriters than are issues without venture backing. This finding is consistent with the notion that higher-quality underwriters are more willing to accept offerings by firms that have been more intensively monitored.

4.2. Ownership position and monitoring by venture capitalists

Table 6 describes the venture capitalists' share ownership and board participation in the companies in which they invest. Venture-capital investors hold significant equity positions in venture-backed firms: on average, they own 34.3% of the pre-IPO equity of the issuer (row 2, table 6). This ownership percentage is higher than those reported in any of the large-blockholder studies cited in table 1 of Jensen and Warner (1988). In one-quarter of the firms, venture capitalists hold at least half the issuer's equity. Typically this stake is shared among two or more venture-capital providers; the mean is 3.0 (row 1). One-quarter of the firms in our sample have only one venture capitalist among their stockholders, and one-quarter have four or more. Such large equity stakes can be expected to give venture capitalists significant incentives to participate in the governance of their portfolio companies.

Even after the IPO, the aggregate equity holdings of venture-capital investors are large, 24.6% on average (row 3). The proportional decline in venture-capital ownership is largely due to the increase in the number of shares in the firm through the offering of new shares, rather than the sale of shares by venture capitalists. On average, venture capitalists sell 6.6% of their pre-IPO shares (row 4), but in 58% of the cases, none of the venture capitalists sell any shares in the offering. Non-venture-capital shareholders sell on average 5.2% of their pre-IPO shares (row 5), a slightly lower percentage (but not significantly lower) than for venture capitalists. Non-venture-capital shareholders are more likely to sell some of their pre-IPO shares, however; only in 44% of the cases do they retain their entire equity stake. A

Table 6
Involvement of venture-capital investors in initial public offerings, 1978–1987.^a

	Variable	Mean	Median	First quartile	Third quartile	Sample size
1.	Number of venture-capital investors per issuer	3.0	2	1	4	433
2.	Pre-IPO equity holdings of all venture-capital investors (% of total)	34.3%	32.5%	16.0%	49.7%	418
3.	Post-IPO equity holdings of all venture-capital investors (% of total)	24.6%	22.6%	11.7%	36.3%	419
4.	Venture capitalists' selling intensity at IPO (number of shares sold by venture-capital investors, divided by their pre-IPO holdings)	6.6%	0.0%	0.0%	9.2%	418
5.	Non-venture-capital insiders' selling intensity at IPO (number of shares sold by non-venture-capital stockholders, divided by their pre-IPO holdings)	5.2%	3.0%	0.0%	9.0%	415
6.	Equity holdings of all venture-capital investors one year after the offering (% of total)	17.8%	14.1%	7.1%	24.5%	301
7.	Number of issuer's board seats held by venture capitalists	2.0	2	1	3	413
8.	Share of issuer's board seats held by venture capitalists (%)	33.4%	28.6%	20.0%	44.4%	410
9.	Number of issuer's board seats held by venture capitalists one year after the offering	1.8	2	1	2	305

^aThe data sources for variables 1–5, 7, and 8 are the 'Management' and 'Selling Stockholders' sections of the IPO prospectus. The data source for variables 6 and 9 is the proxy statement, if any, issued approximately one calendar year after the IPO. We found such proxy statements for 324 of the 433 venture-capital IPOs; the median number of calendar months between the IPO and the proxy date is 13.

chi-square test shows that this difference between venture-capital shareholders and others is significant.

By retaining their share ownership after the offering, the venture capitalists can provide assurance of continued monitoring and can credibly signal their belief in the firm's prospects. Using proxy statements (available for 324 sample firms), we verified that venture capitalists maintain substantial owner-

ship positions in their portfolio companies approximately one year after the IPO. They hold a mean equity share a year after the IPO of 17.8% (row 6), down less than 30% from their immediate post-IPO ownership share of 24.6% (row 3). Thus, contrary to Gladstone's (1989) claim, it is difficult to view the IPO as an immediate 'exit strategy' for the venture-capital providers.

Seats on the board of directors provide one indication of the extent of venture-capital providers' direct participation in the formal control of the companies in which they invest. Venture-capital providers occupy at least one board seat in 369 out of 433 IPO firms, or 85%. Typically venture capitalists hold two seats on the company's board (row 7), or about one-third of all the directors' positions (row 8). This proportion is roughly equivalent to their pre-IPO equity ownership. In 42 IPOs (10% of the total), a venture capitalist chaired the board at the time of the offering. There is very little change in the number of board positions held by venture capitalists one year after the IPO (row 9). These results are consistent with the hypothesis that venture capitalists' large equity holdings are an important source of incentives to monitor and influence management decisions. A similar association between concentrated ownership and board representation is observed in leveraged buyouts [see Baker and Wruck (1989)].

When venture capitalists invest through syndicates, one venture-capital investor typically takes the lead role in assembling the syndicate and working with the portfolio company. This lead venture capitalist has particularly close ties to the IPO company (table 7). Unlike the lead underwriter, the lead venture capitalist is not explicitly identified in the IPO prospectus. Our approach is to define the lead venture capitalist as the venture-capital provider that holds the largest equity position among venture capitalists in a particular IPO and fills a seat on the board. This definition identifies a lead venture capitalist in 288 (66%) of the IPOs. Fourteen of these cases are 'ties', in which two venture capitalists have board seats and hold the same number of shares. In the remaining offerings venture capitalists hold board seats but little equity, or the venture capitalist with the largest equity holdings has no board representation.

The lead venture capitalist is a significant stockholder, owning on average 19.1% of the equity of the firm before the IPO and 13.6% afterward (rows 1 and 2, table 7). Before the offering, the lead venture capitalist serves on the issuer's board for an average of 35.0 months (median = 26 months), or about half the life of the company since incorporation (rows 4 and 5). In fact, 75% of the venture-capital-backed firms had a lead venture capitalist on the board for more than a quarter of their corporate lives.

Like venture capitalists as a group, the lead venture-capital provider sells little of its stock at the IPO and still has a large ownership position a year later (rows 3 and 6). The lead venture capitalists retain, on average, more

Table 7
Involvement of the lead venture-capital investor in initial public offerings, 1978–1987.^a

	Variable	Mean	Median	First quartile	Third quartile	Sample size
1.	Pre-IPO percentage equity holding of lead venture capitalist	19.1%	16.2%	10.4%	23.9%	288
2.	Post-IPO percentage equity holding of lead venture capitalist	13.6%	11.8%	7.8%	17.1%	288
3.	Lead venture capitalist's selling intensity at IPO (number of shares sold by lead venture-capital investor, divided by its pre-IPO holdings)	5.9%	0.0%	0.0%	7.1%	287
4.	Number of months the lead venture capitalist served on the board before the IPO	35.0	26.0	13.0	49.0	266
5.	Percent of firm's life that lead venture capitalist served on the board of directors	51.1%	47.5%	23.5%	85.0%	229
6.	Percent equity holdings of lead venture capitalist one year after the offering	11.6%	9.2%	5.2%	14.9%	204
7.	Dummy variable equal to one if lead venture capitalist still holds its board seat one year after the offering	0.89	1.0	1.0	1.0	223

^aThe data sources for variables 1–5 are the 'Management' and 'Selling Stockholders' sections of the IPO prospectus. The data source for variables 6 and 7 is the proxy statement, if any, issued approximately one calendar year after the IPO. (See note to table 6.)

than 85% of their immediate post-IPO shares a year later – a larger percentage than for venture capitalists overall. In 89% of all IPOs the lead venture capitalist still holds its board seat one year later (row 7).

In summary, we find that venture capitalists are active investors holding large equity positions in the firms in which they invest. The literature on governance of enterprise suggests that high ownership concentration can increase the value of public firms [see Smith (1986) and Wruck (1989)]. Similarly, the presence of active investors with high ownership concentration in leveraged buyouts is associated with increased postbuyout efficiency [see

Kaplan (1989), Muscarella and Vetsuypens (1990), and Smith (1990)]. Our empirical analysis confirms the parallels between leveraged buyout firms and venture-capital providers.

4.3. *Capital markets and the quality of monitoring*

In this section, we examine whether the capital markets recognize variations in the quality of the venture capitalists whose companies are going public. We examine whether monitoring quality is associated with the degree of underpricing of the IPO.

Rock (1986), Beatty and Ritter (1986), and Carter and Manaster (1990) show that the underpricing of IPOs is related to uncertainty about their value. Because underpricing is costly to the issuing firm's pre-IPO shareholders, firms signal their risk by hiring reputable auditors [Titman and Trueman (1986)] and/or underwriters [Beatty and Ritter (1986), Carter and Manaster (1990), and Booth and Smith (1986)].

Similarly, association with skilled venture capitalists may benefit the issuing firm. Because outsiders lack reliable information about the issuer's assets and investment opportunities, differences in the abilities of venture capitalists can provide information to the capital markets. Venture capitalists, with their large equity stakes, have incentives to bring IPOs to market at high values. All else being equal, investors may be willing to pay more for companies brought to market by venture capitalists who are better able to oversee and guide new enterprises. Their ability and monitoring skills should therefore be reflected in a less underpriced issue.

We examine the relation between underpricing and several proxies for the venture capitalist's skills in guiding its portfolio companies:

- (1) The *number of venture capitalists (NVC)* with equity positions in the company before the IPO. The presence of multiple venture capitalists indicates that the issuer has persuaded a larger number of sophisticated investors that the firm has favorable prospects, and that it is willing to open itself up for scrutiny and guidance. Furthermore, the lead venture capitalist has increased incentives to monitor carefully because it has increased the risk to its reputation by soliciting the participation of other venture capitalists. Hence, a larger value for *NVC* should be associated with more intense monitoring, less uncertainty, and, therefore, less underpricing.
- (2) The *length of time that a member of the lead venture capitalist has served on the board of directors (BRD)* of the issuer. The longer the lead venture capitalist has served on the board, the greater its opportunity to observe, oversee, and exercise beneficial influence. Of course, extended board service could instead indicate that the issuer was a marginal company

that required a long time to get to the IPO stage. Nevertheless, given the fact of a public offering, a lengthier working relationship between the venture capitalist and the issuer can reduce underwriters' and investors' uncertainty and, hence, underpricing.

- (3) The *age of the lead venture capitalist* (*AGE*). The longer the lead venture capitalist has been in business, the more experience it should have in monitoring the executives of a startup company and steering the firm onto a healthy course. We expect the variable *AGE* to be negatively associated with uncertainty and with underpricing.
- (4) The *number of prior IPOs* (*DEAL*) in which the lead venture capitalist has participated (since the start of our sample period in January 1978). Since experience with other ventures can be evidence of successful monitoring and guidance, we expect less underpricing for IPOs whose lead venture capitalist has already brought a relatively large number of companies to market.
- (5) The *funds under management* (*CAP*) of the venture capitalist. To attract resources, venture capitalists must convince investors (individuals, corporations, fund managers, financial institutions, etc.) of their expertise and their ability to discover and monitor investment opportunities.⁷ Accordingly, the amount of resources a venture capitalist controls may serve as a proxy for expertise as perceived by investors in venture capital funds. Therefore, we expect less underpricing in IPOs in which large venture capitalists participate.
- (6) The *fraction of the issuing firm's shares owned by the venture capitalists* (*EQ*). The larger the venture capitalists' share of ownership, the stronger their incentives to monitor and participate in management. Further, the degree of control the venture capitalists possess is related to their equity ownership. Thus, we expect larger holdings by venture capitalists to be associated with less underpricing.⁸

Not surprisingly, several of the proxies are correlated in our sample (table 8). The significantly positive correlations among *AGE*, *DEAL*, and *CAP* indicate that older venture capitalists have participated in more IPOs and manage a larger pool of funds than less established venture-capital providers. Also, *EQ*, the fraction of the issuer's shares held by all venture-capital providers, is positively correlated with *NVC*, the number of venture-capital

⁷An exception is the case of captive venture capitalists such as Citicorp Venture Capital, which do not have to compete directly for funds from investors but rather compete with other uses of funds within parent companies.

⁸An alternative variable is the dollar amount invested by venture capitalists in the company. However, our regressions of IPO underpricing use the offer size of the firm as a control variable. To avoid multicollinearity between this control variable and the dollar amount invested in the firm by the venture capitalists, we use the percentage insider holdings by venture capitalists as a measure of the intensity of monitoring by the venture capitalists.

Table 8

Correlation matrix of proxies for quality of venture-capital monitoring, for 433 venture-capital-backed initial public offerings, 1978–1987.^a

	<i>NVC</i>	<i>BRD</i>	<i>AGE</i>	<i>DEAL</i>	<i>CAP</i>
<i>BRD</i>	0.13 ^c				
<i>AGE</i>	−0.01	0.13			
<i>DEAL</i>	0.09	−0.07	0.31 ^b		
<i>CAP</i>	0.05	−0.06	0.46 ^b	0.71 ^b	
<i>EQ</i>	0.55 ^b	0.20 ^b	0.04	0.04	0.02

^a*NVC* = number of venture capitalists with equity in the issuing firm before the IPO.

BRD = number of calendar months between the IPO and the starting date of the lead venture capitalist's board membership.

AGE = number of calendar years between the IPO and the lead venture capitalist's founding year.

DEAL = cumulative number of prior IPOs in which the lead venture capitalist participated.

CAP = capital under management of lead venture capitalist in prior year, scaled by pool of total capital in the venture-capital industry in the same year.

EQ = fraction of issuer's shares held by venture capitalists before the IPO.

^bSignificant at 0.01 level.

^cSignificant at 0.05 level.

shareholders, and with *BRD*, the length of time the lead venture capitalist has served on the issuer's board.

Two or more proxies can highlight different aspects of an IPO. For example, the lead venture capitalist could be a relatively young firm, such as Sevin-Rosen in the Lotus and Compaq IPOs, but with exceptional involvement in the activities of the issuer; Ben Rosen, in this case, served as chairman of the board of Compaq. In 1983, Hambrecht and Quist (H&Q) was the lead venture-capital investor in Information Resources, Inc., which yielded an 87% initial return. H&Q had participated in numerous IPOs by that time and had been an active venture-capital provider for 15 years. Yet, the total equity held by all venture capitalists in Information Resources was a comparatively low 7%, and H&Q's share was much less.

Table 9 reports results of ordinary-least-squares regressions of initial returns against each of the six proxy variables, and against three control variables: aftermarket standard deviation of returns (*SIGMA*), offering size (*SIZE*), and a measure of the lead underwriter's quality. Initial returns are computed as the percentage change from the offering price to the closing price at the end of the first trading day.

Underwriter reputation is measured by four dummy variables, *U1* through *U4*, which represent different levels on the Carter and Manaster (1990) scale, with *U1* representing the lowest-tier underwriters. These variables allow us to compare the relative importance of underwriters and venture capitalists. *SIGMA* has been used extensively in prior empirical work [for instance,

Table 9

Results of OLS regressions of initial returns in initial public offerings against aftermarket standard deviation (*SIGMA*), offer size (*SIZE*), underwriter prestige variables (*UI*, *U2*, *U3*, *U4*), and proxy variables for the monitoring abilities of venture capitalists, 1978–1987 (*t*-statistics in parentheses).^a

	Intercept	<i>SIGMA</i>	<i>SIZE</i>	<i>UI</i>	<i>U2</i>	<i>U3</i>	<i>U4</i>	<i>NVC</i>	<i>BRD</i>	<i>AGE</i>	<i>DEAL</i>	<i>CAP</i>	<i>EQ</i>	<i>N</i>	<i>R</i> ²
1.	0.009 (0.41)	1.687 (3.58) ^c	1.389E-09 (3.03) ^c	-0.052 (-0.66)	0.075 (1.69) ^c	0.016 (0.73)	0.037 (1.97) ^d	-0.008 (-1.86) ^c						395	0.070
2.	-0.001 (-0.04)	1.960 (3.11) ^c	1.716E-09 (3.12) ^c	-0.067 (-0.41)	0.038 (0.58)	0.021 (0.69)	0.046 (1.92) ^c		-0.007 (-1.97) ^d					246	0.111
3.	0.014 (0.36)	2.458 (3.06) ^c	1.591E-09 (2.63) ^c	-0.089 (-0.52)	0.104 (0.84)	0.018 (0.48)	0.031 (1.10)			-0.003 (-2.07) ^d				191	0.116
4.	-0.005 (-0.16)	2.246 (3.07) ^c	1.558E-09 (2.66) ^c	-0.068 (-0.40)	0.125 (1.04)	0.022 (0.62)	0.025 (0.97)				-0.004 (-1.88) ^c			203	0.105
5.	-0.048 (-1.33)	1.994 (2.59) ^d	1.646E-09 (2.57) ^d	— ^b	0.26 (1.75) ^c	0.049 (1.30)	0.048 (1.82) ^c					-0.035 (-0.02)		147	0.118
6.	0.014 (0.58)	1.697 (3.58) ^c	1.271E-09 (2.82) ^c	-0.052 (-0.57)	0.076 (1.73)	0.018 (0.83)	0.038 (2.07) ^d						-0.008 (-2.01) ^d	392	0.071

^a*SIGMA* = aftermarket standard deviation of returns (days 2 through 20).

SIZE = offering size (offer price times number of shares sold).

UI = dummy variable equal to one if Carter–Manaster (1990) underwriter rank falls between 0.0 and 1.5, zero otherwise,

U2 = dummy variable equal to one if Carter–Manaster (1990) underwriter rank falls between 2.0 and 3.5, zero otherwise,

U3 = dummy variable equal to one if Carter–Manaster (1990) underwriter rank falls between 4.0 and 5.5, zero otherwise,

U4 = dummy variable equal to one if Carter–Manaster (1990) underwriter rank falls between 6.0 and 7.5, zero otherwise,

NVC = number of venture capitalists with equity in the issuing firm before the IPO,

BRD = number of calendar months between the IPO and the starting date of the lead venture capitalist's board membership,

AGE = number of calendar years between the IPO and the lead venture capitalist's founding year,

DEAL = cumulative number of prior IPOs in which the lead venture capitalist participated,

CAP = capital under management of lead venture capitalist in prior year, scaled by pool of total capital in the venture-capital industry in the same year,

EQ = fraction of issuer's shares held by venture capitalists before the IPO.

^bNo underwriter had a prestige rank between 0 and 1.5 for the 147 observations for which the variable *CAP* was available. To avoid multicollinearity, variable *UI* was dropped.

^cSignificant at 0.01 level.

^dSignificant at 0.05 level.

^eSignificant at 0.10 level.

Ritter (1984)] to proxy for *ex ante* uncertainty about an issue. These studies generally find a significant positive association between aftermarket volatility and underpricing, which has been interpreted as support for asymmetric information theories of underpricing. We include *SIGMA* in the regressions to aid comparisons with earlier work and to isolate the effect of venture capitalists on uncertainty. Aftermarket volatility is not significantly correlated with any of the venture-capital proxies.

We include the offering size of the IPO (defined as the offer price times the number of shares sold, excluding overallotments) as an additional control variable. Ritter (1984) and Chalk and Peavy (1990) show empirically that larger offerings are less underpriced, although this finding lacks a theoretical explanation. In our sample, by contrast, larger offerings are significantly more underpriced than smaller issues. This surprising finding occurs because our venture-capital-backed sample includes few of the very small offerings that are associated with especially large initial returns. *SIZE* is significantly positively correlated with *NVC*, the number of venture capitalists.

Each of the regressions reported in table 9 shows the dependence of initial returns on the independent variables *SIGMA* and *SIZE*, the underwriter dummy variables *U1* through *U4*, and one of the venture capital proxies. *SIGMA* is significant (at conventional levels) in all regressions, and its coefficients have the expected positive signs. *SIZE* is also significant in all of the regressions.

The coefficients on the underwriter reputation dummy variables are generally insignificant, except occasionally for *U4*, which corresponds to a Carter-Manaster prestige rank between 6.0 and 7.5. (By construction the intercept term captures the effect of the highest-prestige class, a rank between 8.0 and 9.0.) The significant positive coefficient on *U4* in regressions 1, 2, 5, and 6 is consistent with prior findings [for instance, Carter and Manaster (1990) and others] that greater underpricing is associated with less prestigious underwriters.

The coefficients of the proxies for the quality of venture-capital monitoring all have the expected negative sign: superior skills, as suggested by the proxies, are associated with less underpricing. Three proxy variables are significant at the 0.05 level: the length of time the lead venture-capital provider has served on the issuer's board (regression 2), the age of the lead venture capitalist (regression 3), and the percentage of the issuer's equity held by all venture-capital providers (regression 6). Two variables are significant at the 0.10 level: the number of venture capitalists (regression 1) and the number of prior deals in which the lead venture capitalist participated (regression 4). The coefficient on capital under management, however, is statistically insignificant (regression 5). Conceivably the quality of the capital data reported in *Pratt's Guide* is a factor in our failure to detect a significant

relationship.⁹ Although five of the six proxies are significant at the 0.10 level or better when analyzed separately, the venture-capital proxy variables are also significant as a group: a standard *F*-test rejects the null hypothesis that the coefficients on the proxy variables are jointly equal to zero at the 0.05 level.

To summarize, IPO underpricing is related to proxies for the monitoring quality of the issuing firm's venture capitalists. Underpricing is reduced

- (1) the larger the number of venture capitalists owning equity in the issuer,
- (2) the longer the lead venture capitalist has served on the company's board,
- (3) the older the lead venture capitalist is,
- (4) the larger the number of prior IPOs in which the lead venture capitalist participated, and
- (5) the larger the fraction of the issuer's equity owned by venture capitalists.

These results are consistent with the joint hypothesis that the quality of the venture capitalists' monitoring skills reduces investor uncertainty and that lower uncertainty is associated with less IPO underpricing. Coupled with our earlier evidence that venture-capital-backed IPOs are brought to market by higher-quality underwriters, these results show that the quality of the venture capitalists provides incremental information not signaled by the underwriter alone. Monitoring skills and concentrated equity appear to matter to the capital market.

5. Conclusions

This paper is the first detailed empirical investigation of venture-capital investments in enterprises that eventually offer their stock to the public. We find that venture capitalists participate actively in and monitor the progress of their portfolio companies. The venture capitalists specialize in a narrow set of industries for which they can develop expertise; they control concentrated equity positions in their portfolio companies and they serve on the companies' boards.

The venture capitalists continue to hold nearly all of their equity through the offering, and most of their investment a year later. Venture capitalists also generally continue to hold their board positions for more than a year after the IPO. We examine whether the capital markets recognize the monitoring provided by venture capitalists, and find that IPOs with higher-

⁹We compiled information on capital under management from *Pratt's Guide*. The accuracy of this variable is suspect, however, for two reasons. First, many venture capitalists do not report the amount of investment they manage, and it is unclear how this reporting bias affects the data. Second, we frequently encountered entries such as 'in excess of 100 million dollars'. In such cases, we recorded the lower bound described in *Pratt's Guide*, but we do not know by how much we underestimate the true figure.

quality venture capitalists are less underpriced. Taken as a whole, the empirical evidence in this paper shows that venture capitalists, as large blockholders, play an important role in shaping and governing new enterprises.

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